

FRAUDULENT CLAIM DETECTION

Data-Driven Fraud Prevention for Global Insure

Exploratory Data Analysis · Predictive Modelling · Business Strategy

1,000

Claims Analysed

40

Features

~25%

Fraud Rate

84.7%

Model Accuracy

Agenda

End-to-end analysis — from raw data to boardroom recommendations

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Dataset & EDA Overview

1,000 claims · 40 features · data cleaning, class balance, distributions

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EDA Visualizations — Behavior & Claims

Hobby/occupation fraud rates · Claim amount distributions by fraud status

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EDA Visualizations — Geography & Severity

State hotspots · Incident severity paradox · Hour of day · Documentation gaps

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Feature Correlations & Model Evaluation

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Key EDA Insights

Top 6 fraud signals confirmed visually and statistically

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EDA Confirmed by Model

Coefficients & feature importances that validate EDA observations

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Business Recommendations

Area-wise fraud prevention strategy across 4 operational pillars

Dataset & Exploratory Data Analysis

1,000 insurance claims · 40 raw features · target: fraud_reported (Y / N)

1,000

Total Claims

~25%

Fraud Rate

36

Clean Features

10+

New Features Eng.

Data Quality & Cleaning

- ▶ Replaced '?' in collision_type, property_damage, police_report_available
- ▶ Dropped _c39 (100% null), policy_number, incident_location, insured_zip
- ▶ Imputed collision_type (mode) & property_damage (claim-value logic)
- ▶ Removed negative umbrella_limit & illogical premium/claim rows
- ▶ Final: 998 rows × 36 features after cleaning

Class Balance & Splitting Strategy

- ▶ Target: ~75% Legitimate · ~25% Fraudulent (imbalanced)
- ▶ Train / Validation: 70 / 30 stratified split
- ▶ Training set: 698 rows · Validation: 299 rows
- ▶ RandomOverSampler → balanced 1,050 training rows
- ▶ StandardScaler: fitted on train, applied to validation

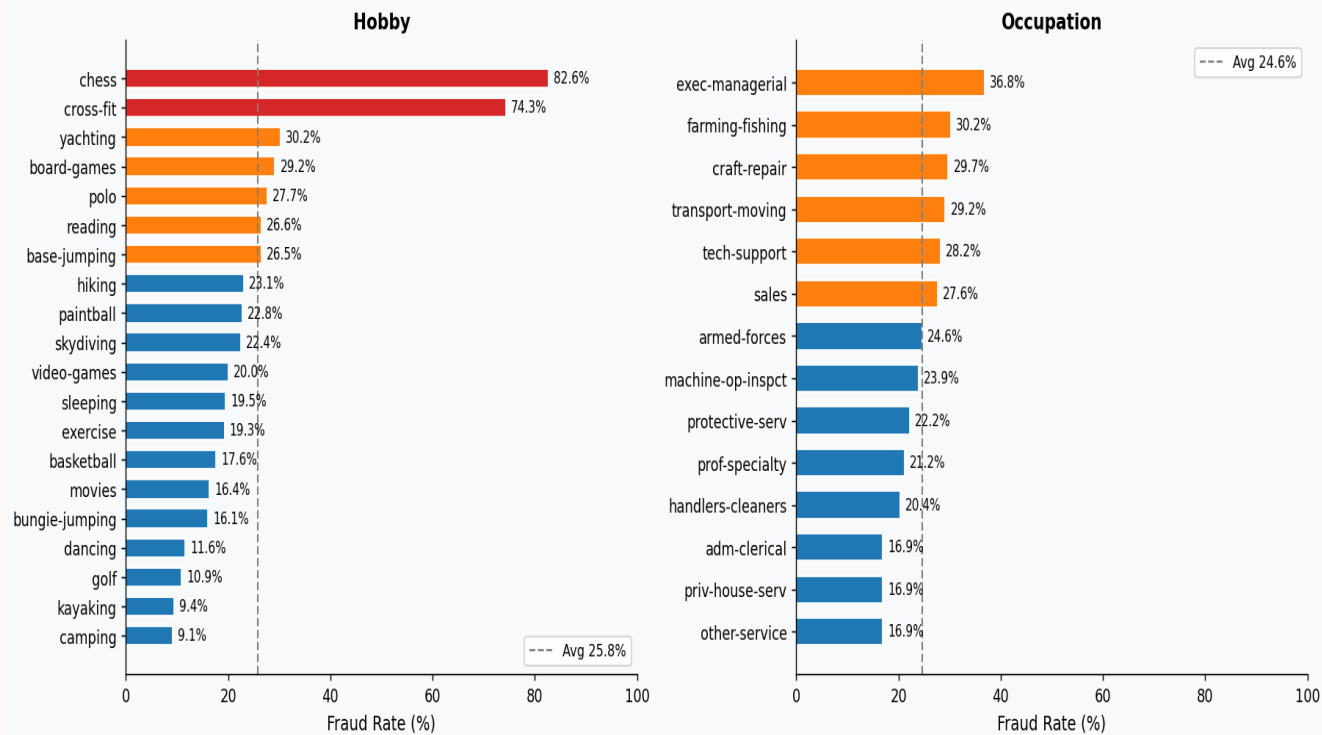
KEY DISTRIBUTION FINDINGS

- ▶ Vehicle & Total Claim → Bimodal distribution (trivial low + genuine mid-range)
- ▶ Incident Hour → Fraud median 13:00 vs Legitimate 11:00 (afternoon spike)
- ▶ Umbrella Limit → Highly right-skewed; most customers skip extra coverage
- ▶ Capital Loss → 50%+ of fraud cases show capital_loss = 0

EDA Visualizations — Behavioral & Claim Patterns

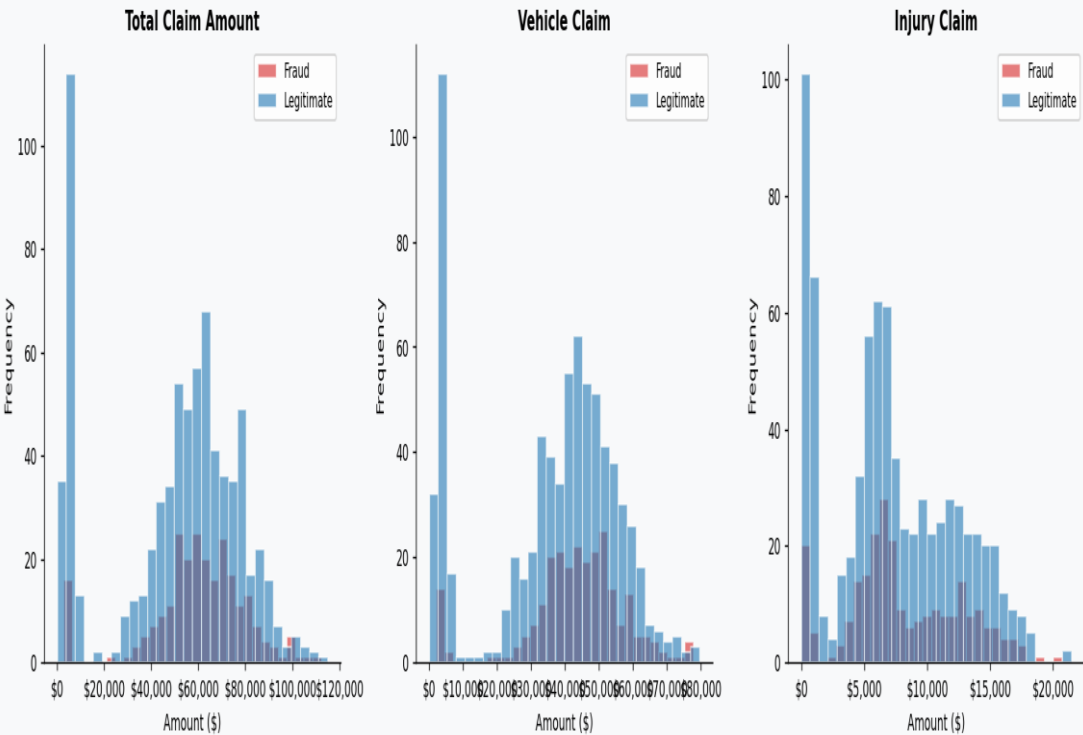
Fraud rate by hobby/occupation · Claim amount distributions by fraud status

Fraud Rate by Hobby & Occupation



chess & cross-fit show ~80% fraud rate · exec-managerial is highest-risk occupation · Red bars exceed the 40% fraud threshold — immediate risk-scoring candidates at policy issuance

Claim Amount Distributions: Fraud vs Legitimate



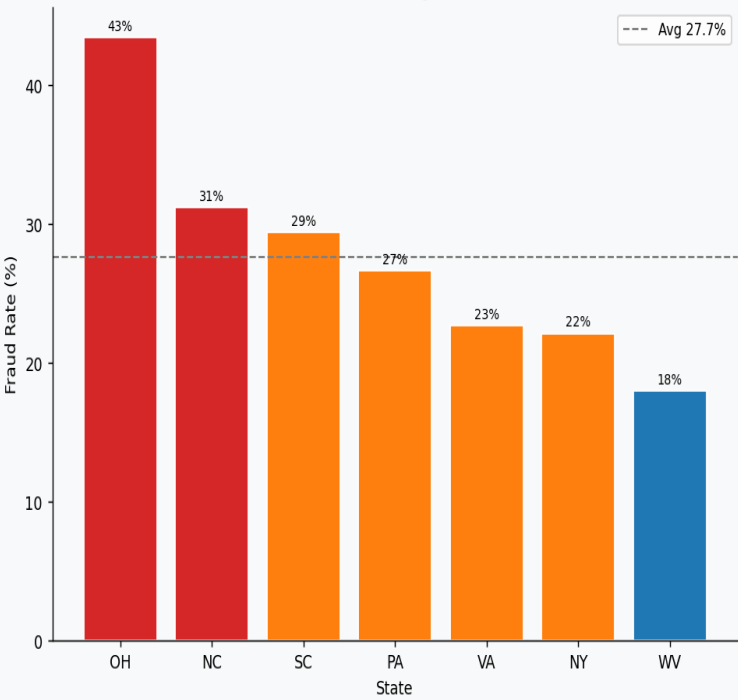
Fraudulent claims cluster at higher vehicle & total claim values · Injury claim shows less separation — fraudsters inflate vehicle damage while minimising body injury claims

EDA Visualizations — Geography, Severity & Documentation

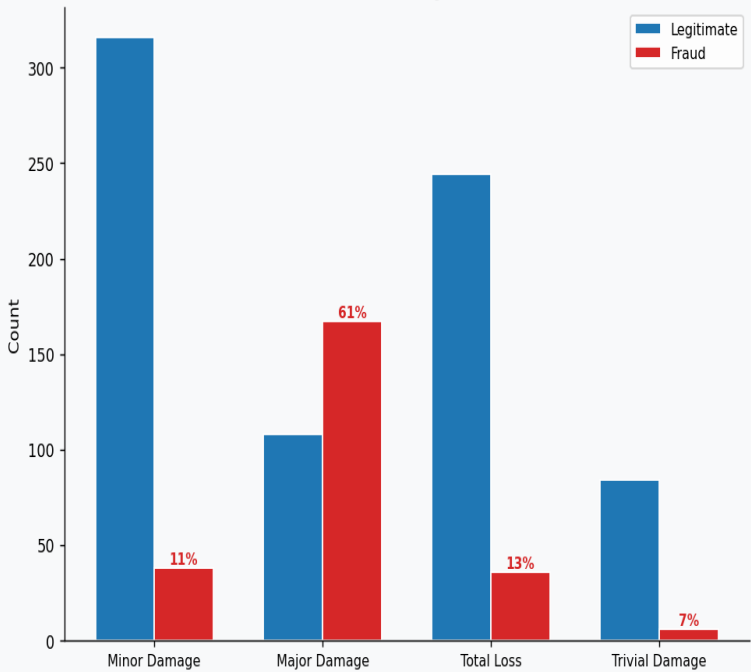
State-level fraud rates · Incident severity paradox · Hour of day & police report patterns

Geographic & Severity Fraud Patterns

Fraud Rate by State



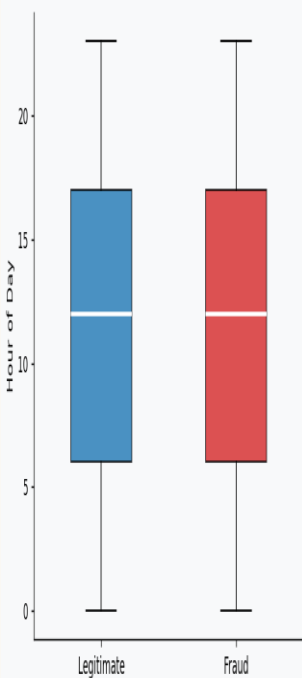
Incident Severity vs Fraud



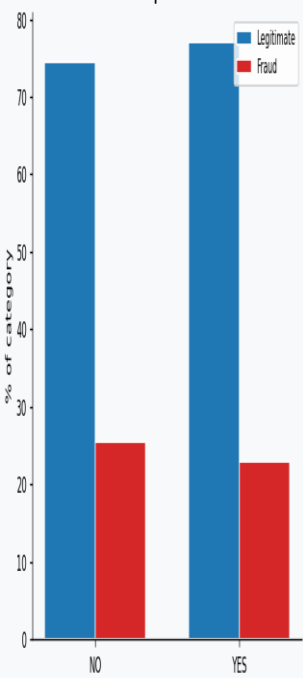
OH & PA are fraud hotspots (>30%) · Major Damage carries the highest fraud % · Fraudsters stage moderate-damage scenes — trivial & total-loss incidents are less likely to be fraudulent

Incident Timing & Documentation Patterns

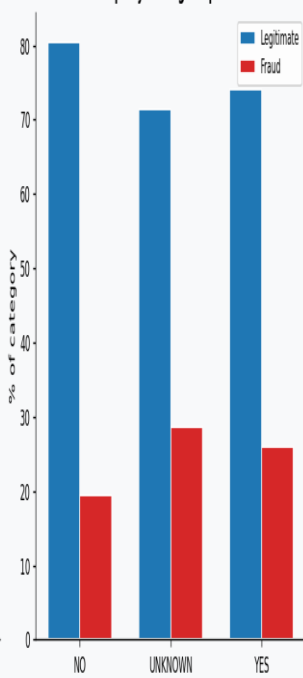
Incident Hour Distribution



Police Report Available



Property Damage Reported

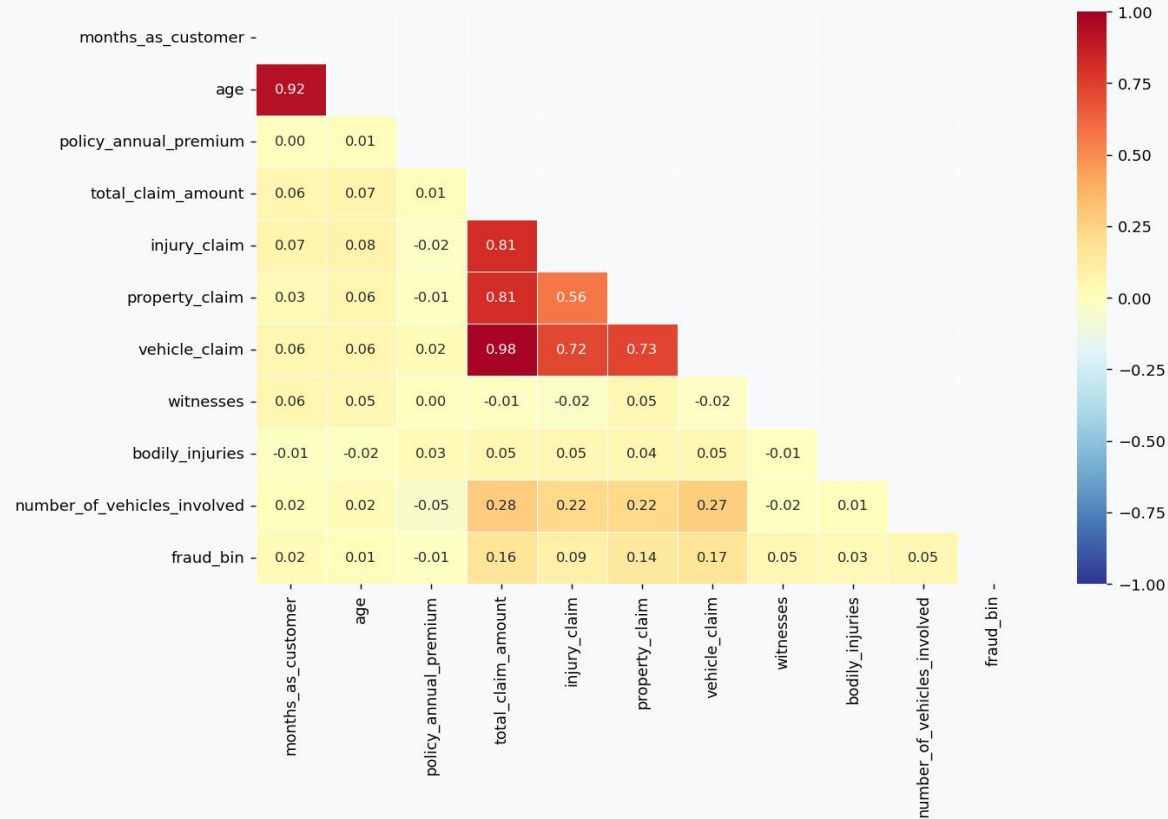


Fraud peaks at hour 13 (vs 11 for legit) · No police report strongly correlates with fraud · Property damage = YES is a significant flag

Feature Correlations & Model Evaluation

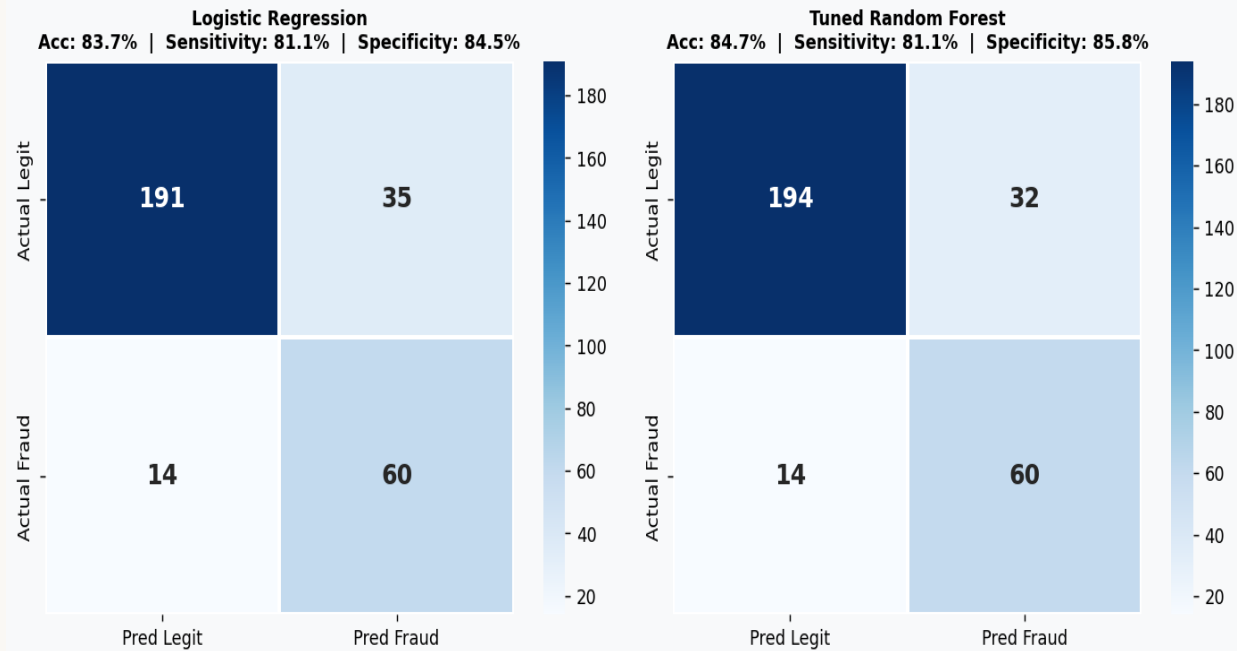
Correlation heatmap for feature selection · Confusion matrices on validation set

Correlation Heatmap — Numerical Features



total_claim · vehicle_claim · injury_claim are highly co-linear → replaced with claim_to_premium_ratio · age & months_as_customer correlated → age dropped to reduce redundancy

Confusion Matrices — Validation Set



Random Forest catches 60/74 fraud cases (81% sensitivity) · 32 false alarms vs 35 for LR · Both models flag ~81% of actual frauds — minimising costly missed detections for the business

Key EDA Insights

Six patterns that clearly distinguish fraudulent from legitimate claims

Hobby Risk Profile

chess & cross-fit → ~80% fraud

Lifestyle signals intent before a claim is filed

Severity Paradox

Major Damage → MORE fraud

Fraudsters stage moderate scenes to appear credible

Vehicle vs Injury Ratio

vehicle_claim >> injury_claim

Staged accidents inflate vehicle damage; minimize injury

Education Skew

Masters / PhD / MD → higher fraud

Higher financial literacy enables sophisticated schemes

Geographic Hotspots

OH & PA → high fraud states

Regional fraud networks require targeted SIU deployment

Witness + No Police Report

≥2 witnesses + no police doc

Pre-arranged witnesses without official records = staging

Model Building & Validation Results

Logistic Regression vs Tuned Random Forest · Stratified 70/30 split

Metric	LR — Train	LR — Validation	RF — Train	RF — Validation
Accuracy	87.62%	83.67%	93.43%	84.67%
Sensitivity	88.57%	81.08%	96.76%	81.08%
Specificity	86.67%	84.51%	90.10%	85.84%
Precision	86.92%	63.16%	90.71%	65.22%
Recall	88.57%	81.08%	96.76%	81.08%
F1-Score	87.74%	71.01%	93.64%	72.29%
AUC-ROC	93.28%	84.24%	98.91%	83.51%

WINNER: Tuned Random Forest — Val. Accuracy 84.67% · F1 72.29% · Best balance of Sensitivity & Specificity

Logistic Regression Strengths

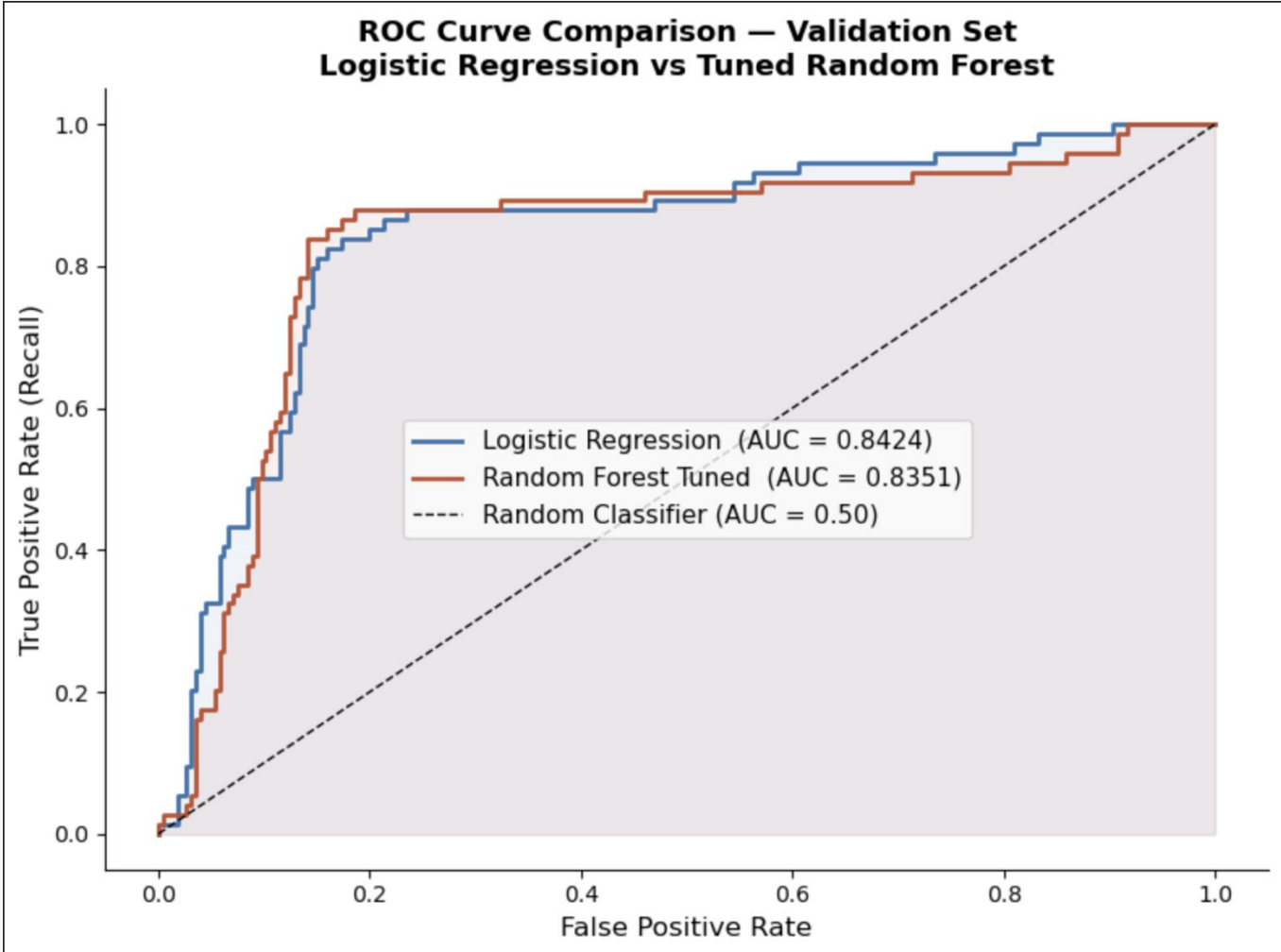
- ▶ 11 features selected by RFECV (optimal subset)
- ▶ AUC-ROC 93.28% — excellent class discrimination
- ▶ Highly interpretable: coefficients map to business rules

Random Forest Strengths

- ▶ Captures non-linear interactions missed by LR
- ▶ GridSearchCV tuned: depth=22, trees=70, leaf=5
- ▶ 5-fold CV accuracy ~92% — robust generalisation
- ▶ Balanced subsample class_weight handles imbalance

ROC-AUC Comparison — Validation Set

Area Under the Curve measures discrimination ability on unseen data · Higher AUC = better model



AUC-ROC SUMMARY

Validation Set (299 records)

Logistic Regression	0.8424
Random Forest (Tuned)	0.8351
Random (baseline)	0.500

What these AUC scores mean

- ▶ AUC 0.8351: RF correctly ranks a random fraud claim above a legit one 83.5% of the time
- ▶ AUC 0.8424: LR is also strong — suitable when interpretability is required
- ▶ Both models far exceed random (0.50), confirming genuine predictive power

EDA Insights Confirmed by the Model

LR coefficients & RF importances validate every major EDA observation

Hobbies: chess & cross-fit

EDA: ~80% fraud rate observed

Model: Coeff +2.14 · $p < 0.001$ · Strongest predictor overall

Vehicle-to-Injury Ratio

EDA: vehicle_claim >> injury_claim in fraud cases

Model: Coeff +0.40 · $p < 0.001$ · Staged accident signal

More Witnesses Present

EDA: fraud cases have higher witness count

Model: Coeff +0.32 · $p = 0.002$ · Organized staging confirmed

Incident Severity

EDA: Major Damage = more fraud

Model: Minor/Trivial/Total Loss all negative (-1.38 to -1.90)

Property Damage = YES

EDA: elevated fraud when property damaged

Model: Coeff +0.40 · $p < 0.001$ · Fully consistent with EDA

Policy CSL 250/500 Limit

EDA: higher coverage tier = higher fraud rate

Model: Coeff +0.47 · $p < 0.001$ · Premium policy fraud magnet

Business Recommendations

Area-wise fraud prevention strategy across four operational pillars

Policy Underwriting

- Risk-score hobbies (chess, cross-fit) at issuance
- Flag PhD / MD / Masters applicants for enhanced checks
- Higher scrutiny for CSL 250/500 policy applications
- 30-day waiting period for high-risk early-claim profiles

Geographic Operations

- Collaborate with local law enforcement in hotspot cities
- Run targeted fraud-awareness campaigns in high-risk states
- Monitor seasonal claim-spike patterns at city level

Claims Processing

- Auto-alert: vehicle_claim > 3× injury_claim
- Mandatory review if claim filed within 60 days of binding
- Flag: ≥2 witnesses + property damage + no police report
- Escalate all 'Major Damage' incidents to SIU automatically